Project Report

Hummel — AI Chat Application

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BS CS - Section: B

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Hummel

Hummel is a mobile AI chat assistant application that allows users to have natural, multi-turn conversations powered by Google Gemini 2.5 Flash. The app is designed with a clean, minimal interface and supports both light and dark modes.

Key Features

- **AI-Powered Conversations:** Real-time chat with Google Gemini 2.5 Flash. The AI responds as "Hummel", a helpful and friendly assistant, with full multi-turn conversation context passed on every request.
- **User Authentication:** Secure email/password registration and login via Firebase Authentication. Users can log out from the Profile screen.
- **Conversation History:** All chats are automatically saved to local storage. The History screen displays conversations grouped by "Today" and "Older", and users can resume any past conversation exactly where they left off.
- **Search Conversations:** A live search bar on the History screen filters saved conversations by their opening message.
- **Edit & Regenerate:** Users can edit any previously sent message to branch the conversation, or tap "Regenerate Response" to get a fresh AI reply to the last message.
- **Stop Generation:** A "Stop generating" button lets users interrupt the AI mid-response.
- **Delete Conversations:** Long-pressing a conversation in the History screen immediately deletes it from both state and local storage.
- **Dark / Light Mode:** A theme toggle in the Profile screen switches between dark and light themes. The selected theme is applied app-wide.
- **Onboarding Flow:** New users are guided through a Splash, Onboarding, Welcome, and Register/Login flow before reaching the main app.
- **Instructions Screen:** Before starting a new chat, users pass through a customisable instructions screen where they can frame or configure their session.

Technologies Used

Framework & Runtime

- **React Native 0.81.5 -** Cross-platform mobile framework for iOS and Android.
- **Expo ~54.0.0 -** Managed workflow toolchain for building and running the app.
- **React 19.1.0 -** UI library powering all component rendering.

Navigation

- **React Navigation -** Stack navigator for screen transitions; Bottom Tab navigator for the main tabs.

State Management

- **Redux Toolkit -** Centralized store for chat messages and conversation data.
- **React Redux -** React bindings for connecting components to the Redux store.

Persistence & Storage

- **AsyncStorage -** Local on-device key-value storage for persisting conversation history.
- **Firebase -** Cloud-based Authentication (email/password sign-up and sign-in).

AI / API

- **Google Gemini 2.5 Flash -** Large language model API used to generate all AI responses.

Theming

- **React Context API -** Custom ThemeContext provides app-wide dark/light mode state and toggle function.

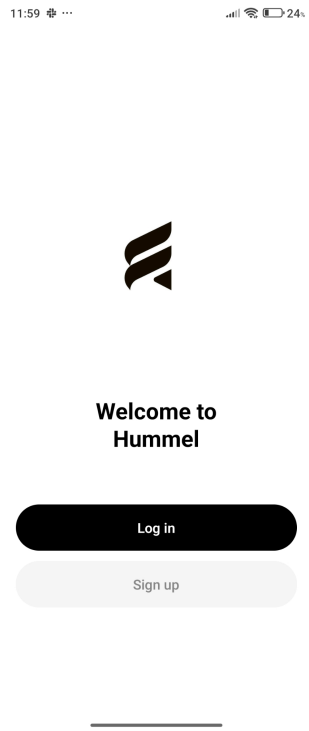
UI Utilities

- **React Native Safe Area Context -** Handles device notches and home-bar insets.
- **React Native Gesture Handler -** Smooth gesture-based interactions and swipe navigation.
- **React Native Screens -** Native screen containers for improved navigation performance.

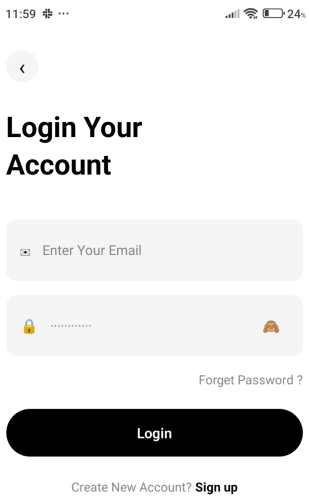
App Screenshots

Splash Screen

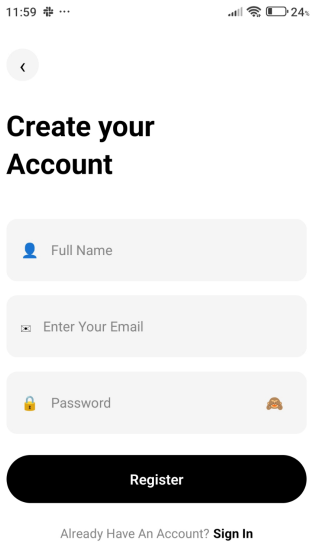
Onboarding Screen



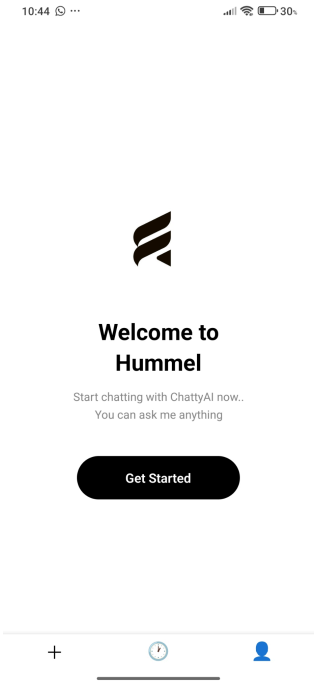
Login Screen



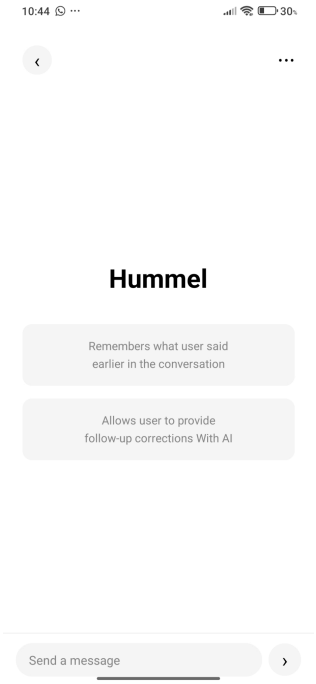
Registration Screen



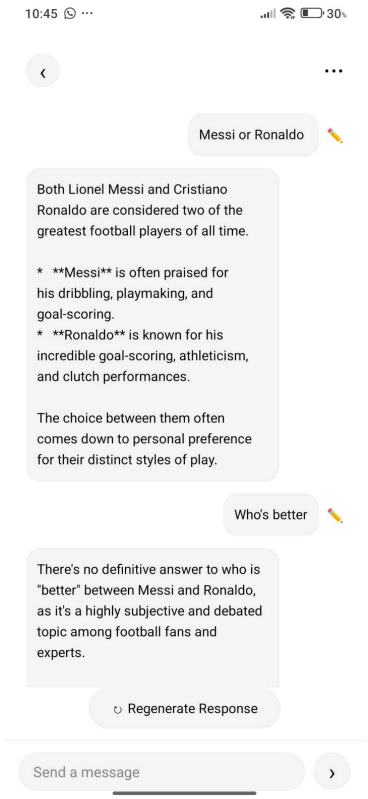
Home Screen (New Chat)

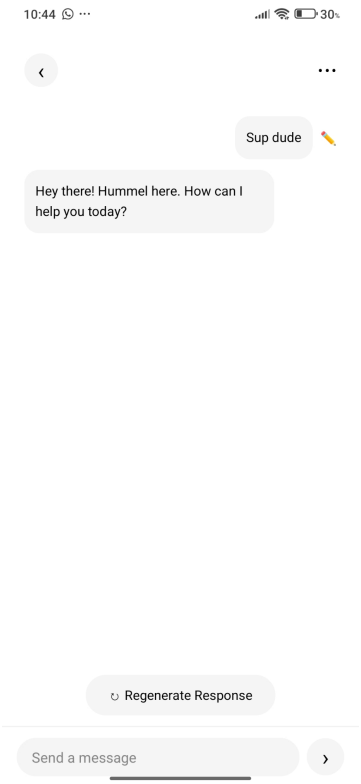


Chat Instructions Screen

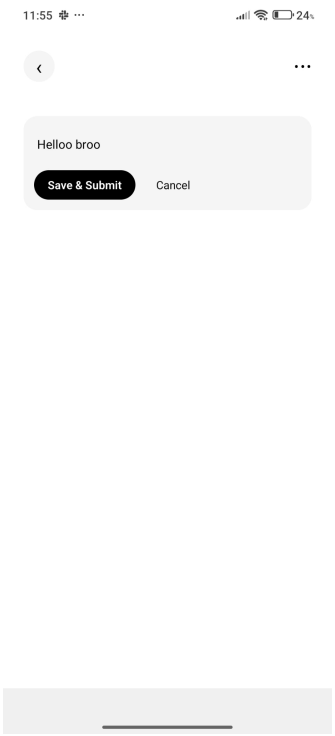


Chat Screen

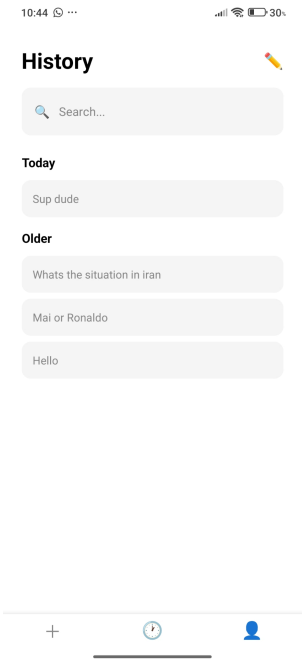
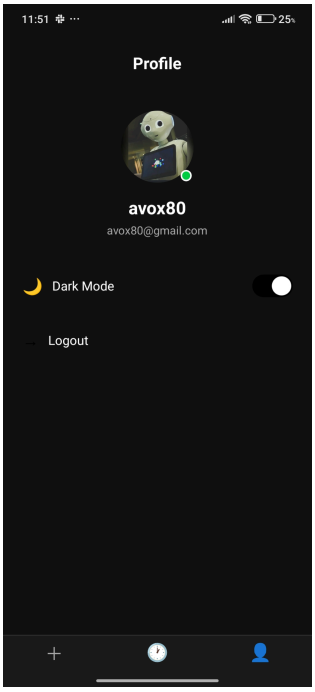
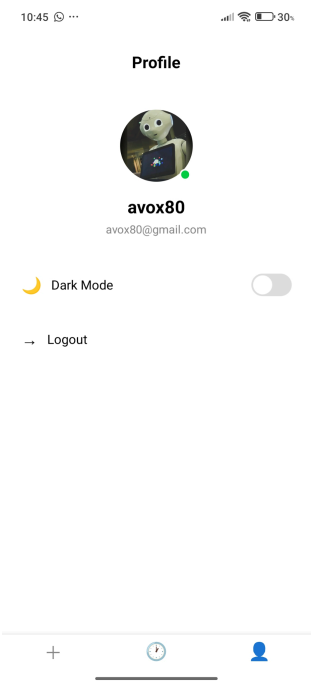




Edit Message Screen



History Screen



Profile Screen (Dark / Light Mode)